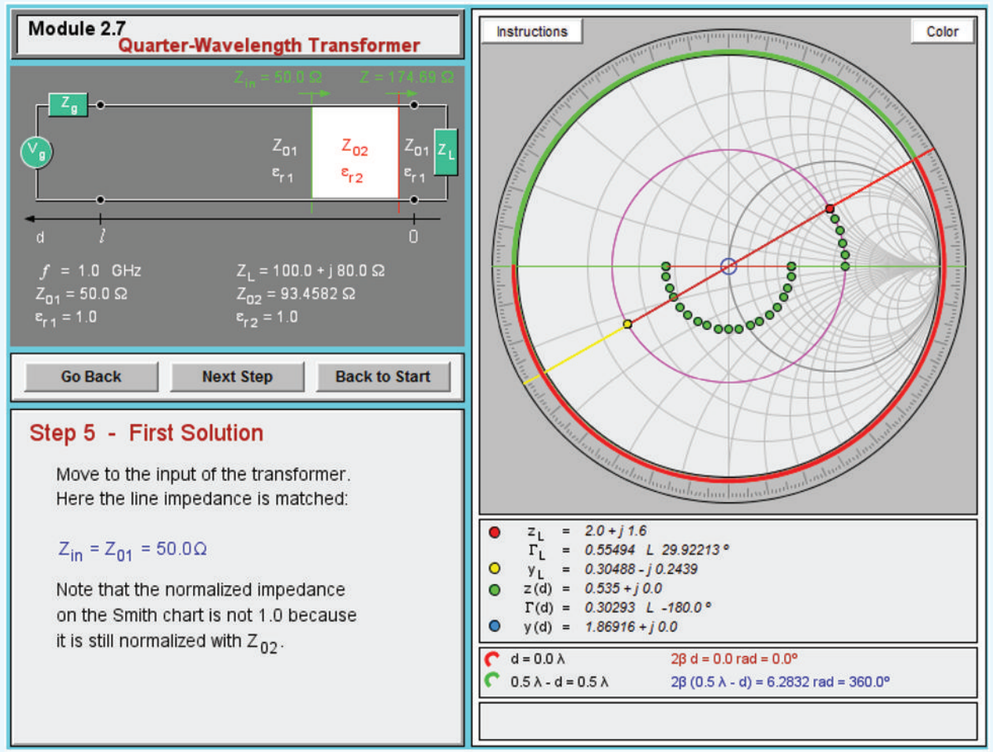
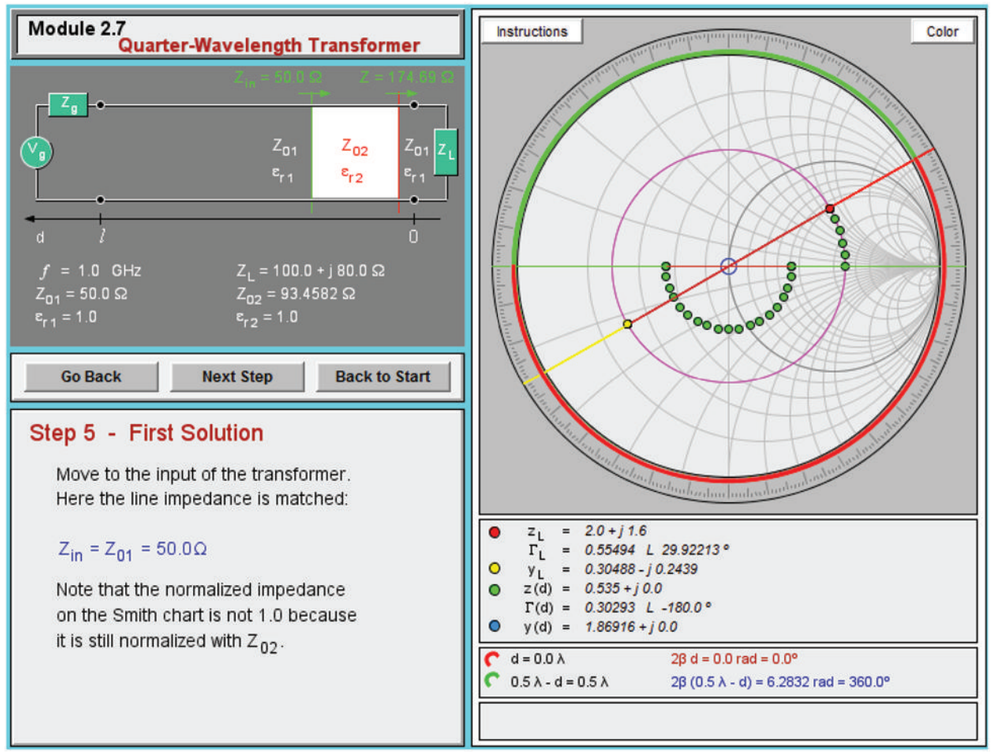
Module 2.7 Quarter-Wavelength Transformer This module allows you to go through a multi-step procedure to design a quarter-wavelength transmission line that, when inserted at the appropriate location on the original line, presents a matched load to the feedline.

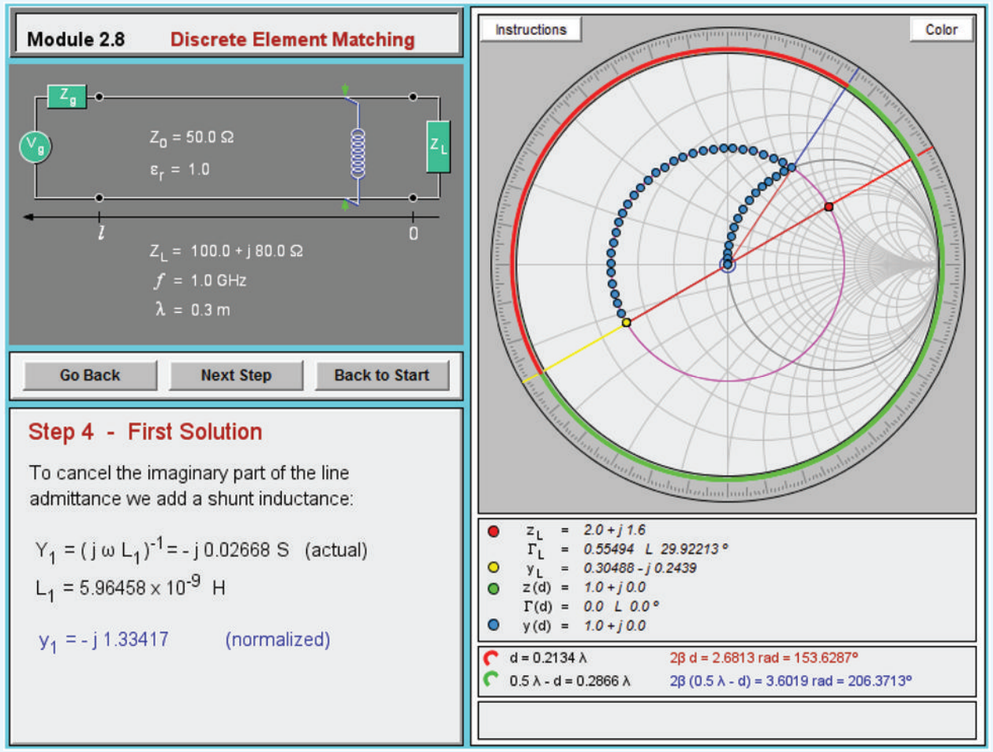


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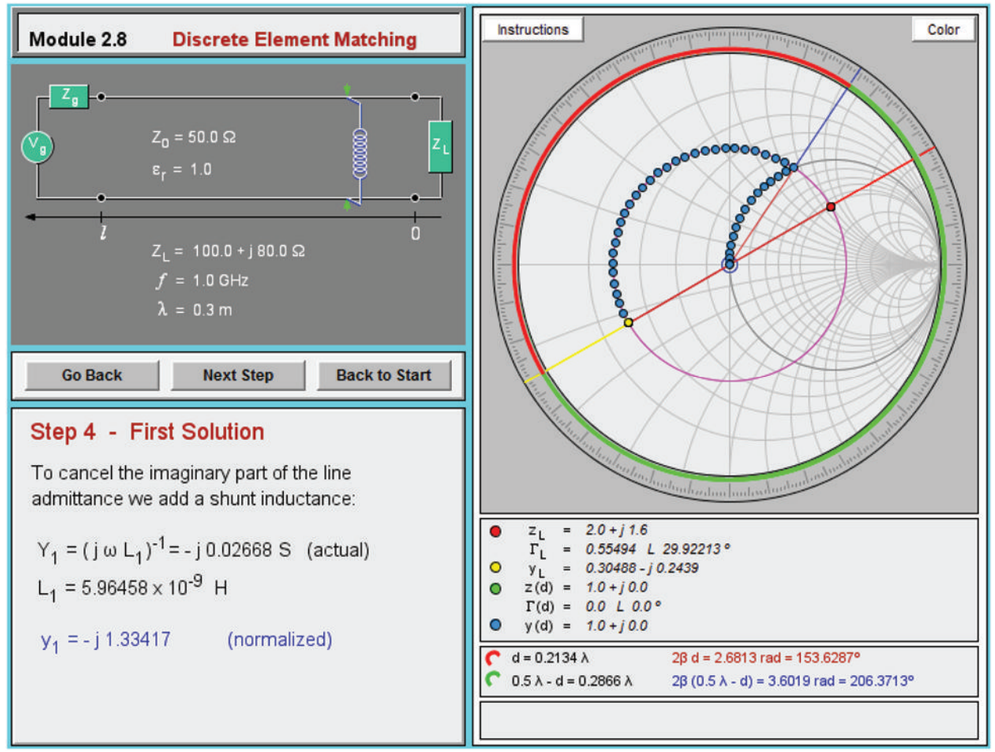
模块 2.7 四分之一波长变压器 该模块允许您通过多步骤程序来设计四分之一波长传输线，当将其插入到原始线路上的适当位置时，可为馈线提供匹配的负载。



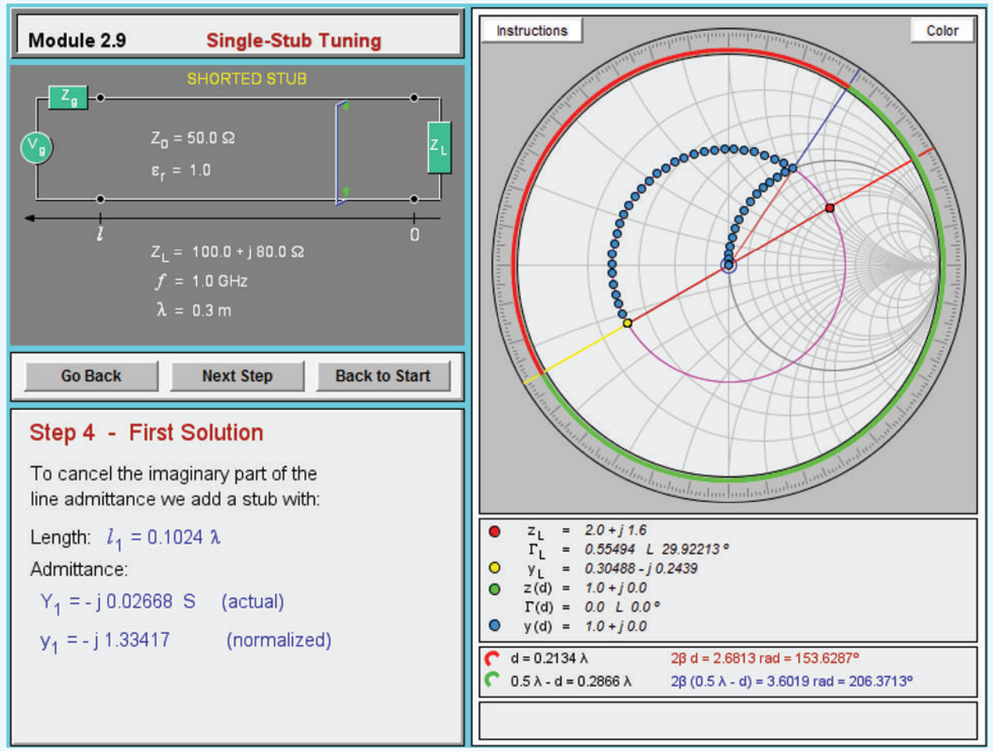
Module 2.8 Discrete Element Matching For each of two possible solutions, the module guides the user through a procedure to match the feedline to the load by inserting a capacitor or an inductor at an appropriate location along the line.



模块 2.8 离散元件匹配 对于两种可能的解决方案中的每一种，该模块都会引导用户完成一个程序，通过在馈线沿线的适当位置插入电容器或电感器来将馈线与负载进行匹配。



Module 2.9 Single-Stub Tuning Instead of inserting a lumped element to match the feedline to the load, this module determines the length of a shorted stub that can accomplish the same goal.



2-12 Transients on Transmission Lines

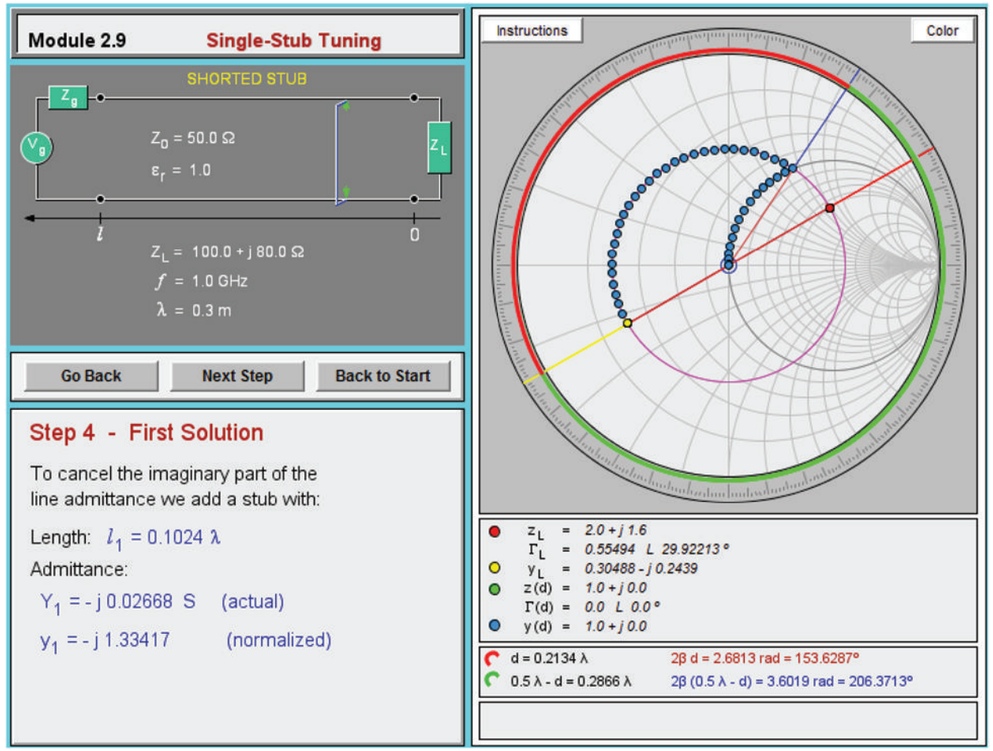
Thus far, our treatment of wave propagation on transmission lines has focused on the analysis of single-frequency, time-harmonic signals under steady-state conditions. The impedance-matching and Smith chart techniques we developed, while useful for a wide range of applications, are inappropriate for dealing with digital or wideband signals that exist in digital chips, circuits, and computer networks. For such signals, we need to examine the transient transmission line response instead.

► The ***transient response*** of a voltage pulse on a transmission line is a time record of its back-and-forth travel between the sending and receiving ends of the line, taking into account all the multiple reflections (echoes) at both ends. ◄

Let us start by considering the case of a single rectangular pulse of amplitude V_0 and duration τ, as shown in **Fig. 2-41(a)**. The amplitude of the pulse is zero prior to $t = 0$, V_0 over the interval $0 \leq t \leq \tau$, and zero afterwards. The pulse can be described mathematically as the sum of two unit step functions:

$$V(t) = V_1(t) + V_2(t) = V_0 u(t) - V_0 u(t - \tau), \qquad (2.143)$$

模块 2.9 单短截线调整 该模块不是插入集总元件来匹配馈线与负载，而是确定可以实现相同目标的短路短截线的长度。



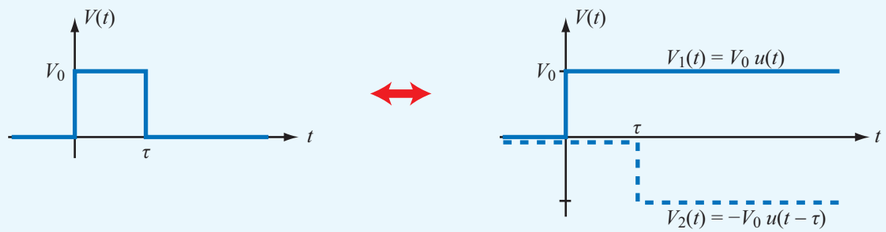
2-12 传输线上的瞬变

到目前为止，我们对传输线上波传播的处理主要集中在稳态条件下单频时谐信号的分析。我们开发的阻抗匹配和史密斯圆图技术虽然可用于广泛的应用，但不适合处理数字芯片、电路和计算机网络中存在的数字或宽带信号。对于此类信号，我们需要检查瞬态传输线响应。

► 传输线上电压脉冲的瞬态响应是其在传输线的发送端和接收端之间来回传输的时间记录，同时考虑了两端的所有多次反射（回声）。◄

让我们首先考虑幅度为 V_0 和持续时间为 τ 的单个矩形脉冲的情况，如图 2-41(a) 所示。脉冲幅度在 $t = 0$ 之前为零，在间隔 $0 \leq t \leq \tau$ 内为 V_0，之后为零。脉冲可以在数学上描述为两个单位阶跃函数的总和：

$$V(t) = V_1(t) + V_2(t) = V_0 u(t) - V_0 u(t - \tau), \qquad (2.143)$$



(a) Pulse of duration τ (b) $V(t) = V_1(t) + V_2(t)$

Figure 2-41 A rectangular pulse $V(t)$ of duration τ can be represented as the sum of two step functions of opposite polarities displaced by τ relative to each other.

where the unit step function $u(x)$ is

$$u(x) = \begin{cases} 1 & \text{for } x > 0, \\ 0 & \text{for } x < 0. \end{cases} \tag{2.144}$$

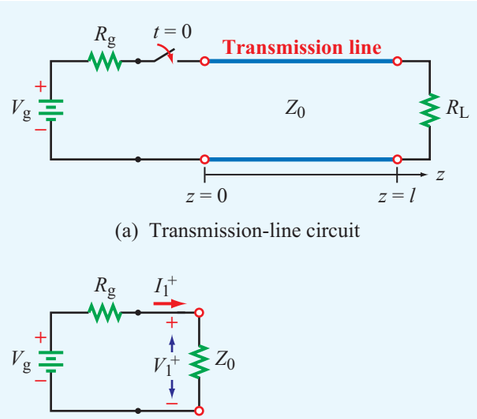
The first component, $V_1(t) = V_0\,u(t)$, represents a dc voltage of amplitude V_0 that is switched on at $t = 0$ and then retains that value indefinitely. The second component, $V_2(t) = -V_0\,u(t-\tau)$, represents a dc voltage of amplitude $-V_0$ that is switched on at $t = \tau$ and remains that way indefinitely. As can be seen from **Fig. 2-41(b)**, the sum $V_1(t) + V_2(t)$ is equal to V_0 for $0 < t < \tau$ and equal to zero for $t < 0$ and $t > \tau$. This representation of a pulse in terms of two step functions allows us to analyze the transient behavior of the pulse on a transmission line as the superposition of two dc signals. Hence, if we can develop basic tools for describing the transient behavior of a single step function, we can apply the same tools for each of the two components of the pulse and then add the results to obtain the response to $V(t)$.

2-12.1 Transient Response to a Step Function

The circuit shown in **Fig. 2-42(a)** consists of a generator, composed of a dc voltage source V_g and a series resistance R_g, connected to a lossless transmission line of length l and characteristic impedance Z_0. The line is terminated in a purely resistive load R_L at $z = l$.

► Note that in previous sections $z = 0$ was defined as the location of the load; now it is more convenient to define it as the location of the source. ◄

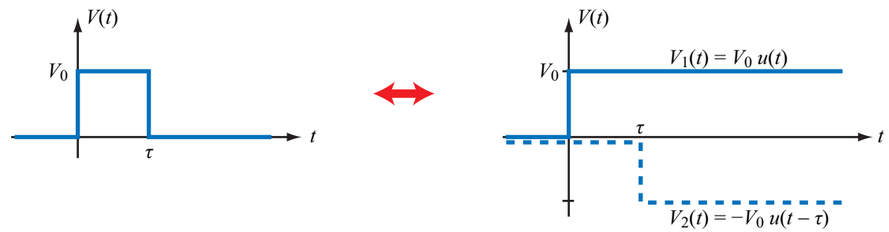
The switch between the generator circuit and the transmission line is closed at $t = 0$. The instant the switch is closed, the transmission line appears to the generator circuit as a load with impedance Z_0. This is because, in the absence of a signal on the line, the input impedance of the line is unaffected by the load impedance R_L. The circuit representing the ***initial condition*** is shown in **Fig. 2-42(b)**. The ***initial current*** I_1^+ and corresponding ***initial voltage*** V_1^+ at the sending end of the



(a) Transmission-line circuit

(b) Equivalent circuit at $t = 0^+$

Figure 2-42 At $t = 0^+$, immediately after closing the switch in the circuit in part (a), the circuit can be represented by the equivalent circuit in part (b). This is because from that instant and until a reflection is received back from the load the generator circuit "sees" only an impedance Z_0.



(a) Pulse of duration τ (b) $V(t) = V_1(t) + V_2(t)$

图 2-41 持续时间为 τ 的矩形脉冲 V(t) 可以表示为两个极性相反的阶跃函数 displa 的和相对于彼此而言τ。

塞德

其中单位阶跃函数 u(x) 为

$$u(x) = \begin{cases} 1 & \text{for } x > 0, \\ 0 & \text{for } x < 0. \end{cases} \tag{2.144}$$

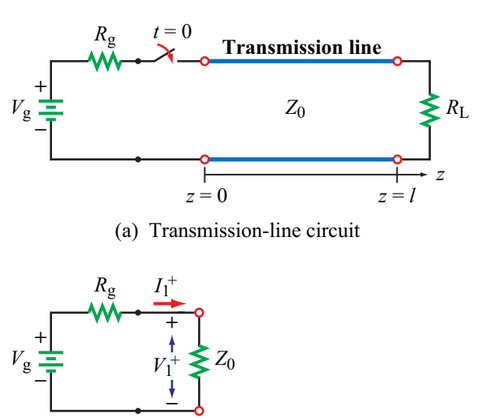
第一个分量 $V_1(t) = V_0\,u(t)$ 表示幅度为 V_0 的直流电压，该电压在 t = 0 处接通，然后无限期地保留该值。第二个分量 $V_2(t) = -V_0\,u(t-\tau)$ 表示幅度为 $-V_0$ 的直流电压，该电压在 $t = \tau$ 处接通并无限期地保持这种状态。从图2-41(b)可以看出，对于$0 < t < \tau$，总和$V_1(t) + V_2(t)$等于V_0，对于$t < 0$和$t > \tau$等于0。这种用两步函数表示脉冲的方式使我们能够将传输线上脉冲的瞬态行为分析为两个直流信号的叠加。因此，如果我们能够开发用于描述单阶函数瞬态行为的基本工具，我们就可以对脉冲的两个分量中的每一个应用相同的工具，然后将结果相加以获得对 V (t) 的响应。

2-12.1 对阶跃函数的瞬态响应

图2-42(a)所示的电路由一个发生器组成，该发生器由直流电压源V_g和串联电阻R_g组成，连接到长度为l、特性阻抗为Z_0的无损传输线。该线路在 z = l 处终止于纯电阻负载R_L。

► 请注意，在前面的章节中，z = 0 被定义为负载的位置；现在将其定义为源的位置会更方便。 ◄

发电机电路和输电线路之间的开关在 t = 0 时闭合。开关闭合的瞬间，



(a) Transmission-line circuit

(b) Equivalent circuit at $t = 0^+$

图2-42 在t=0⁺时，(a)部分电路中的开关闭合后，该电路可以用(b)部分的等效电路表示。这是因为从那一刻起，直到从负载接收到反射为止，发生器电路仅“看到”阻抗 Z_0。

传输线对于发生器电路来说是一个阻抗为 Z_0 的负载。这是因为，当线路上没有信号时，线路的输入阻抗不受负载阻抗 R_L 的影响。代表初始条件的电路如图2-42(b)所示。发送端的初始电流I_1^+和对应的初始电压V_1^+

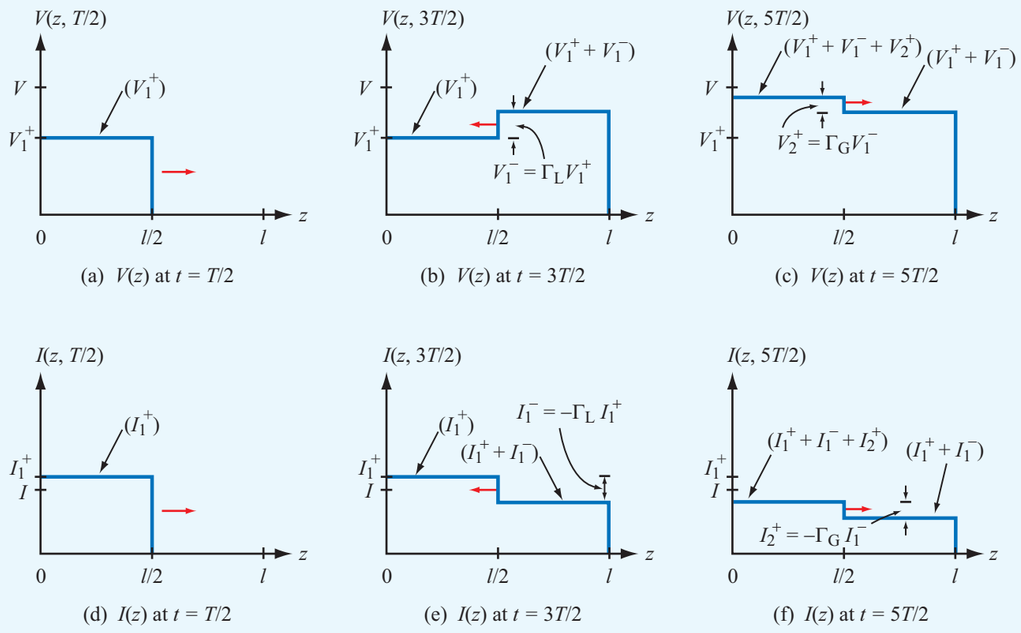


Figure 2-43 Voltage and current distributions on a lossless transmission line at $t = T/2$, $t = 3T/2$, and $t = 5T/2$ due to a unit step voltage applied to a circuit with $R_g = 4Z_0$ and $R_L = 2Z_0$. The corresponding reflection coefficients are $\Gamma_L = 1/3$ and $\Gamma_g = 3/5$.

transmission line are given by

$$I_1^+ = \frac{V_g}{R_g + Z_0}, \tag{2.145a}$$

$$V_1^+ = I_1^+ Z_0 = \frac{V_g Z_0}{R_g + Z_0}. \tag{2.145b}$$

The combination of V_1^+ and I_1^+ constitutes a wave that travels along the line with velocity $u_p = 1/\sqrt{\mu\varepsilon}$ immediately after the switch is closed. The plus-sign superscript denotes the fact that the wave is traveling in the $+z$ direction. The transient response of the wave is shown in **Fig. 2-43** at each of three instances in time for a circuit with $R_g = 4Z_0$ and $R_L = 2Z_0$. The first response is at time $t_1 = T/2$, where $T = l/u_p$ is the time it takes the wave to travel the full length of the line. By time t_1, the wave has traveled halfway down the line; consequently, the voltage on the first half of the line is equal to V_1^+, while the voltage on the second half is still zero (**Fig. 2-43(a)**). At $t = T$, the wave reaches the load at $z = l$, and because $R_L \neq Z_0$, the mismatch generates a reflected wave with amplitude

$$V_1^- = \Gamma_L V_1^+, \tag{2.146}$$

where

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0} \tag{2.147}$$

is the reflection coefficient of the load. For the specific case illustrated in **Fig. 2-43**, $R_L = 2Z_0$, which leads to $\Gamma_L = 1/3$. After this first reflection, the voltage on the line consists of the sum of two waves: the initial wave V_1^+ and the reflected wave V_1^-. The voltage on the transmission line at $t_2 = 3T/2$ is shown in **Fig. 2-43(b)**; $V(z, 3T/2)$ equals V_1^+ on the first half of the line ($0 \le z < l/2$) and $(V_1^+ + V_1^-)$ on the second half ($l/2 \le z \le l$).

At $t = 2T$, the reflected wave V_1^- arrives at the sending end of the line. If $R_g \neq Z_0$, the mismatch at the sending end generates a reflection at $z = 0$ in the form of a wave with

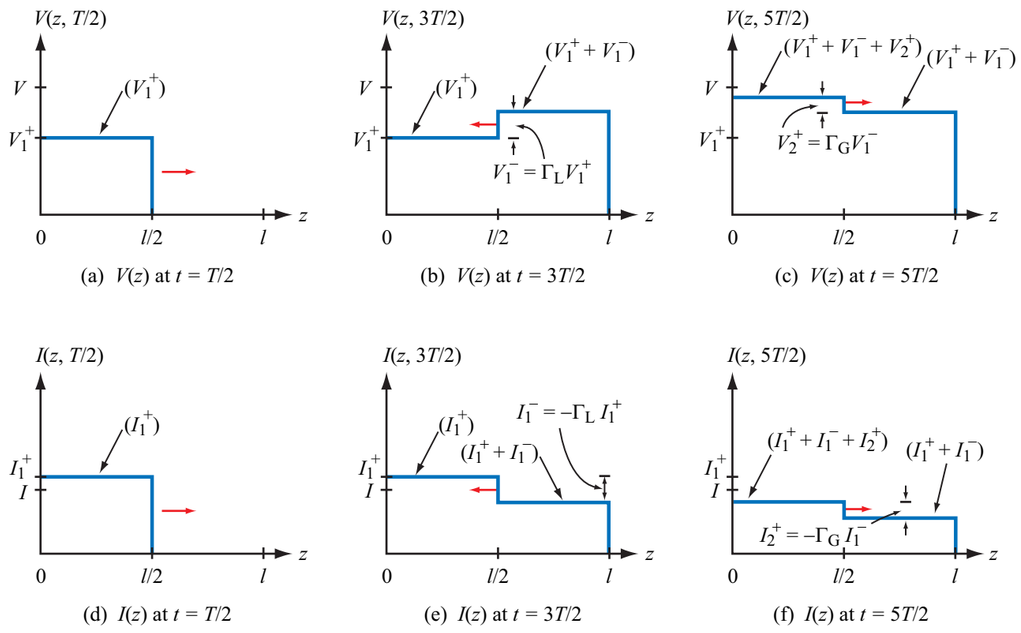


图 2-43 由于将单位阶跃电压施加到具有 $R_g = 4Z_0$ 和 $R_L = 2Z_0$ 的电路，无损传输线上在 t = T /2、t = 3T /2 和 t = 5 T /2 处的电压和电流分布。相应的反射系数为 $\Gamma_L = 1/3$ 和 $\Gamma_g = 3/5$。

传输线由下式给出

$$I_1^+ = \frac{V_g}{R_g + Z_0}, \tag{2.145a}$$

$$V_1^+ = I_1^+ Z_0 = \frac{V_g Z_0}{R_g + Z_0}. \tag{2.145b}$$

V_1^+ 和 I_1^+ 的组合构成了一个波，在开关闭合后立即以速度 $u_p = 1/\sqrt{\mu\varepsilon}$ 沿直线传播。加号上标表示波沿 +z 方向传播。对于具有 $R_g = 4Z_0$ 和 $R_L = 2Z_0$ 的电路，波的瞬态响应如图 2-43 所示。第一个响应出现在时间 $t_1 = T/2$，其中 $T = l/u_p$ 是波走完全程所用的时间。到时间 t_1 时，波已沿线行进了一半；因此，线路前半部分的电压等于 V_1^+，而后半部分的电压仍然为零（图2-43（a））。在 t = T 处，波到达 z = l 处的负载，并且因为 $R_L \neq Z_0$，

不匹配会产生具有振幅的反射波

$$V_1^- = \Gamma_L V_1^+, \tag{2.146}$$

在哪里

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0} \tag{2.147}$$

是负载的反射系数。对于图 2-43 所示的具体情况，$R_L = 2Z_0$，导致 $\Gamma_L = 1/3$。经过第一次反射后，线路上的电压由两个波之和组成：初始波 V_1^+ 和反射波 V_1^-。$t_2 = 3T/2$时传输线上的电压如图2-43(b)所示； $V(z, 3T/2)$ 等于该行前半部分 $(0 \le z < l/2)$ 上的 V_1^+ 和后半部分 $(l/2 \le z \le l)$ 上的 $(V_1^+ + V_1^-)$。

在 t = 2T 时，反射波 V_1^- 到达线路的发送端。如果 $R_g \neq Z_0$，发送端的不匹配会在 z = 0 处以波的形式产生反射